Thermodynamics

It is the only physical theory of universal content concerning which I am convinced that, within the framework of the applicability of its basic concepts, it will never be overthrown.

What this unit sets out to do

- explain the terms system and surroundings, discriminate between close, open and isolated systems, and explain internal energy, work and heat
- state the first law of thermodynamics and express it mathematically, calculate energy changes as work and heat contributions in chemical systems, explain the state functions U and H, correlate and measure them, and define standard states
- calculate enthalpy changes for various types of reactions, state and apply Hess's law of constant heat summation, and differentiate between extensive and intensive properties
- define spontaneous and non-spontaneous processes, explain entropy as a thermodynamic state function and apply it for spontaneity, explain Gibbs energy change, and establish the relationship between it and spontaneity and the equilibrium constant

Why energy changes are worth a unit of their own

- chemical energy stored by molecules can be released as heat during chemical reactions when a fuel like methane, cooking gas or coal burns in air, and it may also be used to do mechanical work when a fuel burns in an engine, or to provide electrical energy through a galvanic cell like a dry cell
- various forms of energy are interrelated and under certain conditions may be transformed from one form into another, and the study of these energy transformations forms the subject matter of thermodynamics
- the laws of thermodynamics deal with energy changes of macroscopic systems involving a large number of molecules rather than microscopic systems containing a few, and they are not concerned with how or at what rate the transformations are carried out but are based on the initial and final states of a system undergoing the change, applying only when a system is in equilibrium or moves from one equilibrium state to another, where macroscopic properties like pressure and temperature do not change with time
- the questions this unit answers are how we determine the energy changes involved in a chemical reaction or process, whether it will occur or not, what drives it, and to what extent chemical reactions proceed

In this unit

5.1 Thermodynamic Terms

5.2 Applications

5.3 Measurement of ΔU and ΔH

5.4 Enthalpy Change of a Reaction

5.5 Enthalpies for Different Reactions

5.6 Spontaneity

5.7 Gibbs Energy and Equilibrium

A beaker holding the reacting substances, which is the system

The room around it, which is the surroundings

A wall between them, real or imaginary, which is the boundary

Figure 5.A System and the surroundings

Open, matter and energy both crossing the boundary

Closed, energy crossing but not matter

Isolated, neither crossing

Figure 5.B Open, closed and isolated systems.

5.1 Thermodynamic terms, and 5.1.1 the system and the surroundings

Where the system ends

- a system in thermodynamics refers to that part of the universe in which observations are made, and the remaining universe constitutes the surroundings, the two together constituting the universe
- the entire universe other than the system is not affected by the changes taking place in the system, so for all practical purposes the surroundings are that portion of the remaining universe which can interact with it, usually the region of space in its neighbourhood
- the system may be defined by physical boundaries like a beaker or test tube, or simply by a set of Cartesian coordinates specifying a particular volume in space, and it is necessary to think of it as separated from the surroundings by some sort of wall, real or imaginary, called the boundary, which is designed to let us control and keep track of all movements of matter and energy in or out

5.1.2 Types of the system

Open, closed and isolated

- in an open system there is exchange of energy and matter between system and surroundings, the presence of reactants in an open beaker being an example, where the boundary is an imaginary surface enclosing the beaker and reactants
- in a closed system there is no exchange of matter but exchange of energy is possible, the presence of reactants in a closed vessel made of conducting material such as copper or steel being an example

A vessel whose wall lets no heat pass either way

The wall is called an adiabatic wall, and the system an adiabatic one

Any change made in such a system is an adiabatic process

Figure 5.C An adiabatic system which does not permit the transfer of heat through its boundary.

Isolated systems, and the state of a system

- in an isolated system there is no exchange of energy or matter between the system and the surroundings, the presence of reactants in a thermos flask or any other closed insulated vessel being an example
- the system must be described by specifying quantitatively each of its properties, such as pressure, volume and temperature, as well as its composition, and it must be described before and after the change
- the state of a thermodynamic system is described by its measurable or macroscopic properties, so we can describe the state of a gas by quoting its pressure, volume, temperature and amount, and variables like p, V and T are called state variables or state functions because their values depend only on the state of the system and not on how it is reached
- it is not necessary to define all the properties to define the state completely, since only a certain number can be varied independently, that number depending on the nature of the system, and once those minimum macroscopic properties are fixed the others automatically have definite values, while the state of the surroundings can never be completely specified, and fortunately it is not necessary to do so

5.1.4 The internal energy as a state function

What internal energy is, and what changes it

- when we talk about our chemical system losing or gaining energy we need a quantity representing the total energy of the system, which may be chemical, electrical, mechanical or any other type, the sum of all of them being what thermodynamics calls the internal energy U
- it may change when heat passes into or out of the system, when work is done on or by the system, and when matter enters or leaves the system
- to examine a change in internal energy by doing work, take a system containing some water in a thermos flask or an insulated beaker, which allows no exchange of heat through its boundary, calling the initial state A at temperature TA with internal energy UA

A vessel of water with thermally conducting walls, made of copper

The whole enclosed in a large heat reservoir held at a different temperature

Heat crosses the boundary, so the change in temperature measures the heat absorbed

Figure 5.D A system which allows heat transfer through its boundary.

Two ways of doing the same work

- one way is to do some mechanical work, say 1 kJ, by rotating a set of small paddles and churning the water, calling the new state B at temperature TB, where it is found that TB is greater than TA, the change in internal energy being UB minus UA
- the second way is to do an equal amount of electrical work with an immersion rod, and the change in temperature is found to be the same as in the first case
- experiments in this manner were done by J.P. Joule between 1840 and 1850, and he showed that a given amount of work done on the system, no matter how it was done and irrespective of path, produced the same change of state as measured by the change in temperature

So U is a state function

- it is appropriate to define a quantity, the internal energy U, whose value is characteristic of the state of a system, such that the adiabatic work required to bring about a change of state equals the difference between the value of U in one state and in the other
- by the conventions of IUPAC in chemical thermodynamics, the positive sign expresses that the adiabatic work is positive when work is done on the system and the internal energy increases, and it will be negative if the work is done by the system, since then the internal energy decreases
- V, p and T are other familiar state functions, so if we change the temperature from 25 degrees C to 35 degrees C the change is plus 10 degrees whether we go straight up or cool the system first and then heat it, and the volume of water in a pond is a state function because the change in volume is independent of whether the water came by rain or by tubewell or by both

Heat

- we can also change the internal energy of a system by transfer of heat from the surroundings to the system or the other way, without expenditure of work, and this exchange of energy, which is a result of a temperature difference, is called heat, q
- taking water at temperature TA in a container with thermally conducting walls, say of copper, enclosed in a huge heat reservoir at temperature TB, the heat absorbed by the system can be measured in terms of the temperature difference, and here the change in internal energy equals q, since no work is done at constant volume
- by the conventions of IUPAC, q is positive when heat is transferred from the surroundings to the system

Pressure plotted against volume for a compression in one step

The external pressure held constant throughout

The work done on the gas is the shaded rectangle under that line

Figure 5.E Work done on an ideal gas in a cylinder when it is compressed by a constant external pressure, pex (in single step) is equal to the shaded area.

The general case, and the first law

- in the general case a change of state is brought about both by doing work and by transfer of heat, and the change in internal energy is then $\Delta U = q + w$
- for a given change in state q and w can vary depending on how the change is carried out, however their sum depends only on the initial and final states and is independent of the way the change is carried out, and if there is no transfer of energy as heat or as work, that is in an isolated system, then ΔU is zero
- $\Delta U = q + w$ is the mathematical statement of the first law of thermodynamics, that the energy of an isolated system is constant, commonly stated as the law of conservation of energy, that energy can neither be created nor destroyed
- there is considerable difference between the character of the thermodynamic property energy and that of a mechanical property such as volume, since we can specify an unambiguous absolute value for the volume of a system in a particular state but not for the internal energy, of which we can measure only the changes

Problem 5.1, and 5.2 applications

- the question asks for the change in internal energy when no heat is absorbed but work is done on the system, when no work is done but q is taken out of the system, and when w is done by the system while q is supplied to it
- in the first case ΔU equals the adiabatic work and the wall is adiabatic, in the second ΔU equals minus q and the walls are thermally conducting, and in the third ΔU equals q minus w and it is a closed system
- many chemical reactions involve the generation of gases capable of doing mechanical work, or the generation of heat, and it is important to quantify these changes and relate them to the changes in internal energy

5.2.1 Work

Pressure-volume work

- we consider only mechanical work, that is pressure-volume work, taking a cylinder containing one mole of an ideal gas fitted with a frictionless piston, of total volume Vi and internal pressure p
- if the external pressure is greater than p the piston moves inward until the pressure inside becomes equal to it

Pressure against volume for a compression carried out in a few finite steps

The external pressure held constant within each step and raised between them

The work done on the gas is the shaded staircase of rectangles

Figure 5.F pV-plot when pressure is not constant and changes in finite steps during compression from initial volume, Vi to final volume, Vf . Work done on the gas is represented by the shaded area.

The same compression carried out in infinitely many infinitesimal steps

The external pressure only ever a hair above the pressure of the gas

The staircase becomes a smooth curve, and the work is the area beneath it

Figure 5.G pV-plot when pressure is not constant and changes in infinite steps (reversible conditions) during compression from initial volume, Vi to final volume, Vf . Work done on the gas is represented by the shaded area.

Working the expression out

- if the change is achieved in a single step to a final volume Vf, and during the compression the piston moves a distance l across a cross-sectional area A, then the volume change is l times A, that is Vf minus Vi
- since pressure is force per unit area, the force on the piston is the external pressure times A, so the work done on the system is force times distance, which comes to minus the external pressure times the change in volume
- the negative sign is required to obtain the conventional sign for w, which will be positive, indicating that in compression work is done on the system, since Vf minus Vi is negative and negative multiplied by negative is positive
- if the pressure is not constant at every stage but changes in a number of finite steps, the work done on the gas is summed over all the steps

Reversible processes

- if the pressure changes during the process such that it is always infinitesimally greater than the pressure of the gas, then at each stage the volume decreases by an infinitesimal amount and the work is found by integrating the external pressure over the volume change
- in compression the external pressure at each stage equals the internal pressure plus an infinitesimal amount, and in an expansion under similar conditions it is always less than the pressure of the system, such processes being called reversible processes
- a process or change is said to be reversible if it is brought about in such a way that it could at any moment be reversed by an infinitesimal change, so a reversible process proceeds infinitely slowly through a series of equilibrium states, the system and surroundings always in near equilibrium with each other

Relating work to the internal pressure

- processes other than reversible processes are known as irreversible processes
- under reversible conditions the external pressure at each stage is the internal pressure plus an infinitesimal amount, and since the product of those infinitesimals is very small, the reversible work is minus the integral of the internal pressure over the volume change
- the pressure of the gas can then be expressed in terms of its volume through the gas equation, so for n moles of an ideal gas at constant temperature the reversible work comes to minus 2.303 nRT times the logarithm of Vf over Vi, and for an adiabatic change, where q is zero, ΔU equals the adiabatic work

Free expansion, and the forms of the first law

- expansion of a gas into vacuum is called free expansion, and no work is done during the free expansion of an ideal gas whether the process is reversible or irreversible
- substituting the work term into the first law gives ΔU = q minus the external pressure times the change in volume, so if a process is carried out at constant volume then ΔU equals qV, the subscript denoting that heat is supplied at constant volume
- for isothermal expansion of an ideal gas into vacuum w is zero since the external pressure is zero, and Joule determined experimentally that q is zero too, so ΔU is zero, while for an isothermal irreversible change q equals minus w equals the external pressure times the volume change, and for an isothermal reversible change q equals minus w equals nRT times the logarithm of Vf over Vi

Problems 5.2, 5.3 and 5.4

- Problem 5.2 takes two litres of an ideal gas at 10 atm expanding isothermally at 25 degrees C into a vacuum until its total volume is 10 litres, and since the external pressure is zero, q equals minus w equals zero, so no work is done and no heat is absorbed
- Problem 5.3 takes the same expansion against a constant external pressure of 1 atm, giving q equals minus w equals 8 litre-atm
- Problem 5.4 takes the expansion of Problem 5.2 for 1 mol of an ideal gas conducted reversibly, giving q equals minus w equals 2.303 nRT log of Vf over Vi, which works out to 393.66 L atm

5.2.2 Enthalpy, H

Why another state function was needed

- the heat absorbed at constant volume equals the change in internal energy, but most chemical reactions are carried out not at constant volume but in flasks or test tubes under constant atmospheric pressure, so we need another state function suitable for those conditions
- at constant pressure the first law reads ΔU = qp minus pΔV, where qp is the heat absorbed by the system and the second term represents the expansion work done by the system
- writing the initial state as 1 and the final as 2 and rearranging gives qp as the difference between U2 plus pV2 and U1 plus pV1

The definition, and why it is a state function

- we define another thermodynamic function, the enthalpy H, from the Greek enthalpien, to warm or heat content, as H = U + pV, so that qp equals H2 minus H1, that is ΔH
- although q is a path dependent function, H is a state function because it depends on U, p and V, all of which are state functions, so ΔH is independent of path and hence qp is also independent of path
- for finite changes at constant pressure ΔH = ΔU + pΔV, and it is important to note that when heat is absorbed by the system at constant pressure we are actually measuring changes in the enthalpy
- ΔH is negative for exothermic reactions, which evolve heat during the reaction, and positive for endothermic reactions, which absorb heat from the surroundings, while at constant volume ΔH equals ΔU equals qV

When the difference between ΔH and ΔU matters

- the difference is not usually significant for systems consisting of only solids or liquids, since these do not suffer any significant volume changes upon heating, but it becomes significant when gases are involved
- using the ideal gas law for the gaseous reactants and products at constant pressure and temperature gives pΔV equal to Δng RT, where Δng is the number of moles of gaseous products minus the number of moles of gaseous reactants
- substituting that gives ΔH = ΔU + Δng RT, which is useful for calculating ΔH from ΔU and the other way round

A container of gas at volume V and temperature T

Then the same container divided down the middle by a partition

Each half now holds half the volume, but the temperature in each is still T

Figure 5.H A gas at volume V and temperature T

Problem 5.5

- if water vapour is assumed to be a perfect gas, the molar enthalpy change for vapourisation of 1 mol of water at 1 bar and 100 degrees C is 41 kJ mol-1, and the question asks for the internal energy change
- for the change from liquid to gas, ΔU equals ΔH minus Δng RT, and substituting gives 41.00 kJ mol-1 minus 1 times 8.3 J mol-1 K-1 times 373 K, that is 41.00 minus 3.096, which is 37.904 kJ mol-1

Extensive and intensive properties

- an extensive property is one whose value depends on the quantity or size of matter present in the system, for example mass, volume, internal energy, enthalpy and heat capacity
- those properties which do not depend on the quantity or size of matter present are known as intensive properties, for example temperature, density and pressure
- a molar property is the value of an extensive property of the system for 1 mol of the substance, so dividing by the amount of matter makes it independent of that amount, other examples being molar volume and molar heat capacity
- taking a gas in a container of volume V at temperature T and partitioning it so the volume is halved, each part now has half the original volume but the temperature remains T, which makes it clear that volume is an extensive property and temperature an intensive one

Heat capacity

- heat transferred to a system appears as a rise in its temperature, and the increase of temperature is proportional to the heat transferred
- the magnitude of the coefficient depends on the size, composition and nature of the system, so we write q = C ΔT, where C is the heat capacity

A sealed steel vessel, the bomb, holding the sample in pure dioxygen

The bomb sitting in a water bath so no heat escapes to the surroundings

The volume cannot change, so no work is done and the heat measured is qV

Figure 5.I Bomb calorimeter

Specific heat, and Cp against CV

- the specific heat is the quantity of heat required to raise the temperature of one unit mass of a substance by one degree celsius, or one kelvin, so the heat required for a sample is the specific heat times the mass times the temperature change
- at constant volume the heat capacity is denoted CV and at constant pressure Cp, so that qV equals CV ΔT equals ΔU, and qp equals Cp ΔT equals ΔH
- for a mole of an ideal gas ΔH = ΔU + Δ(pV), which is ΔU + RΔT, and putting the two heat capacities into that gives Cp minus CV equals R

5.3 Measurement of ΔU and ΔH, calorimetry

How the measurement is made

- we can measure energy changes associated with chemical or physical processes by an experimental technique called calorimetry, in which the process is carried out in a vessel called a calorimeter immersed in a known volume of liquid
- knowing the heat capacity of that liquid and of the calorimeter, it is possible to determine the heat evolved by measuring temperature changes, and measurements are made under two different conditions, at constant volume and at constant pressure
- heat absorbed at constant volume is measured in a bomb calorimeter, where a steel vessel, the bomb, is immersed in a water bath so that no heat is lost to the surroundings, a combustible substance being burnt in pure dioxygen supplied in the bomb and the heat evolved transferred to the surrounding water whose temperature is monitored
- since the bomb is sealed its volume does not change, so the energy changes are measured at constant volume and no work is done, even for reactions involving gases, and the temperature change is then converted to qV using the known heat capacity of the calorimeter

An insulated vessel open to the atmosphere, so the pressure stays constant

The reaction mixture inside, with a thermometer and a stirrer

The heat measured at constant pressure is qp, which is the enthalpy of reaction

Figure 5.J Calorimeter for measuring heat changes at constant pressure (atmospheric pressure).

Measuring ΔH

- measurement of heat change at constant pressure, generally under atmospheric pressure, can be done in a calorimeter open to the air
- since ΔH equals qp at constant pressure, the heat absorbed or evolved at constant pressure is also called the heat of reaction or enthalpy of reaction
- in an exothermic reaction heat is evolved and the system loses heat to the surroundings, so qp is negative and the reaction enthalpy is also negative, while in an endothermic reaction heat is absorbed, qp is positive and the reaction enthalpy is positive

Problem 5.6

- 1 g of graphite is burnt in a bomb calorimeter in excess of oxygen at 298 K and 1 atmospheric pressure, the temperature rising from 298 K to 299 K, and with the heat capacity of the calorimeter at 20.7 kJ per K the question asks for the enthalpy change
- the quantity of heat absorbed by the calorimeter is its heat capacity times the temperature change, and the heat from the reaction has the same magnitude but opposite sign, since the heat lost by the reaction mixture equals the heat gained by the calorimeter, giving minus 20.7 kJ, the negative sign indicating the exothermic nature of the reaction
- so ΔU for the combustion of 1 g of graphite is minus 20.7 kJ, and for 1 mol it is 12.0 g mol-1 times that over 1 g, which is minus 2.48 × 102 kJ mol-1, and since Δng is zero, ΔH equals ΔU equals minus 2.48 × 102 kJ mol-1

5.4 Enthalpy change of a reaction, reaction enthalpy

What reaction enthalpy is

- in a chemical reaction reactants are converted into products, and the enthalpy change accompanying the reaction is called the reaction enthalpy
- it is the sum of the enthalpies of the products minus the sum of the enthalpies of the reactants

Each substance with its melting point and its boiling point

Beside them the standard enthalpy of fusion and of vaporisation

The stronger the intermolecular forces, the larger both values run

Figure 5.K Standard Enthalpy Changes of Fusion and Vaporisation (Tf and Tb are melting and boiling points, respectively)

Reaction enthalpy from molar enthalpies

- the reaction enthalpy is the sum over the products of the stoichiometric coefficient times the molar enthalpy, minus the same sum over the reactants, the coefficients being those in the balanced chemical equation
- so for methane burning to carbon dioxide and water it is the molar enthalpy of CO2 plus twice that of liquid water, minus the molar enthalpy of CH4 plus twice that of O2
- knowledge of this quantity is required when one needs to plan the heating or cooling to maintain an industrial chemical reaction at constant temperature, and to calculate the temperature dependence of the equilibrium constant

Standard states, and phase transformations

- enthalpy of a reaction depends on the conditions under which it is carried out, so standard conditions must be specified, the standard enthalpy of reaction being the enthalpy change when all the participating substances are in their standard states
- the standard state of a substance at a specified temperature is its pure form at 1 bar, so the standard state of liquid ethanol at 298 K is pure liquid ethanol at 1 bar and that of solid iron at 500 K is pure iron at 1 bar, and usually data are taken at 298 K, standard conditions being denoted by a superscript on the symbol
- phase transformations also involve energy changes, ice for example requiring heat for melting, normally at constant atmospheric pressure and at a constant temperature of 273 K, the enthalpy of fusion of water being 6.00 kJ mol-1, and if water freezes the process is reversed and an equal amount of heat is given off to the surroundings
- the enthalpy change accompanying the melting of one mole of a solid in its standard state is called the standard enthalpy of fusion or molar enthalpy of fusion, and since melting is endothermic all enthalpies of fusion are positive

Vaporisation and sublimation

- the amount of heat required to vaporize one mole of a liquid at constant temperature and under standard pressure is called its standard enthalpy of vaporization or molar enthalpy of vaporization, that of water being plus 40.79 kJ mol-1
- sublimation is the direct conversion of a solid into its vapour, so solid CO2 or dry ice sublimes at 195 K with an enthalpy of 25.2 kJ mol-1 and naphthalene sublimes slowly with 73.0 kJ mol-1, the standard enthalpy of sublimation being the change in enthalpy when one mole of a solid sublimes at constant temperature under standard pressure
- the magnitude of the enthalpy change depends on the strength of the intermolecular interactions in the substance, so the strong hydrogen bonds between water molecules hold them tightly in the liquid phase whereas in an organic liquid such as acetone the dipole-dipole interactions are significantly weaker, and it takes less heat to vaporise 1 mol of acetone than 1 mol of water

Problem 5.7, the swimmer's film of water

- a swimmer coming out of a pool is covered with a film of water weighing about 18 g, and the question asks how much heat must be supplied to evaporate it at 298 K and for the internal energy of vaporisation, given the standard enthalpy of vaporisation of water at 298 K as 44.01 kJ mol-1
- 18 g of water is 1 mol, so the heat supplied is 1 mol times 44.01 kJ mol-1, that is 44.01 kJ
- assuming steam behaves as an ideal gas, the internal energy of vaporisation is the enthalpy minus Δng RT, that is 44.01 kJ minus 2.48 kJ, which is 41.53 kJ

Problem 5.8

- assuming water vapour to be a perfect gas, the question asks for the internal energy change when 1 mol of water at 100 degrees C and 1 bar is converted to ice at 0 degrees C
- the data given are the enthalpy of fusion of ice at 6.00 kJ mol-1 and the heat capacity of water at 4.2 J per g per degree C

Selected substances, each with the standard enthalpy of its formation at 298 K

The value is for one mole formed from the elements in their reference states

An element in its own reference state is entered as zero by convention

Figure 5.L Standard Molar Enthalpies of Formation (Δf H) at 298K of a Few Selected Substances

The end of Problem 5.8

- the enthalpy change comes to minus 7.56 kJ mol-1 plus minus 6.00 kJ mol-1, that is minus 13.56 kJ mol-1
- there is negligible change in volume during the change from liquid to solid, so pΔV is zero and ΔH equals ΔU equals minus 13.56 kJ mol-1

Standard enthalpy of formation

- the standard enthalpy change for the formation of one mole of a compound from its elements in their most stable states of aggregation, also known as reference states, is called the standard molar enthalpy of formation, the subscript f indicating that one mole of the compound has been formed in its standard state from its elements in their most stable states
- the reference state of an element is its most stable state of aggregation at 25 degrees C and 1 bar, so the reference state of dihydrogen is H2 gas and those of dioxygen, carbon and sulphur are O2 gas, graphite and rhombic sulphur respectively

What counts as a formation enthalpy, and what does not

- the formation of liquid water from H2 and O2 has ΔfH of minus 285.8 kJ mol-1, that of methane from graphite and hydrogen minus 74.81 kJ mol-1, and that of liquid ethanol minus 277.7 kJ mol-1
- a standard molar enthalpy of formation is a special case of a reaction enthalpy, where one mole of a compound is formed from its constituent elements, so the exothermic reaction of CaO with CO2 to give CaCO3 at minus 178.3 kJ mol-1 is not an enthalpy of formation, since calcium carbonate has been formed from other compounds and not from its elements
- nor is the reaction of H2 with liquid Br2 to give 2HBr an enthalpy of formation, since two moles instead of one are formed, so dividing all coefficients by 2 gives the enthalpy of formation of HBr gas as minus 36.4 kJ mol-1
- by convention the standard enthalpy of formation of an element in its reference state, that is its most stable state of aggregation, is taken as zero

Working out a decomposition

- to find how much heat is required to decompose calcium carbonate to lime and carbon dioxide with all substances in their standard states, the reaction enthalpy is the sum over products of coefficient times formation enthalpy, minus the same sum over reactants
- here both coefficients are 1, so it is the formation enthalpy of CaO plus that of CO2 minus that of CaCO3, which is minus 635.1 plus minus 393.5 minus minus 1206.9, giving 178.3 kJ mol-1
- so the decomposition of solid CaCO3 is an endothermic process and it has to be heated to get the desired products

Thermochemical equations

- a balanced chemical equation together with the value of its reaction enthalpy is called a thermochemical equation, and we specify the physical state, along with the allotropic state, of each substance in it, as in the combustion of liquid ethanol at minus 1367 kJ mol-1, whose negative sign indicates an exothermic reaction
- the coefficients in a balanced thermochemical equation refer to the number of moles, never molecules, of reactants and products involved
- the numerical value of the standard reaction enthalpy refers to the number of moles specified by the equation, and it has units of kJ mol-1

Iron oxide reduced by hydrogen

- to calculate the heat of reaction for solid Fe2O3 with 3H2 gas giving 2Fe solid and 3H2O liquid, we take the standard formation enthalpies from the table
- those are minus 285.83 kJ mol-1 for liquid water and minus 824.2 kJ mol-1 for solid Fe2O3, while for solid Fe and for H2 gas the value is zero by convention
- so the reaction enthalpy is 3 times minus 285.83 minus 1 times minus 824.2, that is minus 857.5 plus 824.2, which comes to minus 33.3 kJ mol-1

What a mole of reaction means

- the coefficients used in these calculations are pure numbers equal to the respective stoichiometric coefficients
- the unit kJ mol-1 means per mole of reaction, and once we balance the chemical equation in a particular way, that defines the mole of reaction

Hess's law of constant heat summation

- adding the equation for the combustion of graphite to carbon dioxide and the reverse of the combustion of carbon monoxide gives the formation of CO from graphite and half an O2, for which the reaction enthalpy is minus 393.5 plus 283.0, that is minus 110.5 kJ mol-1
- in general, if the enthalpy of an overall reaction from A to B along one route is ΔrH, and there are enthalpies for a series of reactions leading to the same product B along another route, then ΔrH is the sum of those

5.5 Enthalpies for different types of reactions

Standard enthalpy of combustion

- combustion reactions are exothermic in nature and are important in industry, rocketry and other walks of life, and the standard enthalpy of combustion is the enthalpy change per mole of a substance when it undergoes combustion, all the reactants and products being in their standard states at the specified temperature
- cooking gas in cylinders contains mostly butane, and during complete combustion of one mole of butane 2658 kJ of heat is released, while combustion of glucose gives out 2802.0 kJ per mole
- our body also generates energy from food by the same overall process as combustion, although the final products are produced after a series of complex bio-chemical reactions involving enzymes

Problem 5.9, the enthalpy of formation of benzene

- the combustion of one mole of benzene takes place at 298 K and 1 atm, producing CO2 gas and liquid water and liberating 3267.0 kJ of heat, and the question asks for the standard enthalpy of formation of benzene, given the formation enthalpies of CO2 gas and liquid water as minus 393.5 and minus 285.83 kJ mol-1
- multiplying the formation of CO2 by 6 gives minus 2361 kJ mol-1 and the formation of water by 3 gives minus 857.49 kJ mol-1, and summing them gives minus 3218.49 kJ mol-1
- reversing the combustion equation gives plus 3267.0 kJ mol-1, and adding that to the sum gives the formation of liquid benzene from 6 graphite and 3H2 at plus 48.51 kJ mol-1

What atomization means

- taking the atomization of dihydrogen, H2 gas going to 2H gas with an enthalpy of 435.0 kJ mol-1, the H atoms are formed by breaking H-H bonds
- the enthalpy change in this process is known as the enthalpy of atomization, the enthalpy change on breaking one mole of bonds completely to obtain atoms in the gas phase
- in the case of diatomic molecules like dihydrogen the enthalpy of atomization is also the bond dissociation enthalpy, and for a metal that sublimes to a monatomic vapour it is the same as the enthalpy of sublimation

The route drawn out

- the whole of this page carries the working for the combustion of graphite to carbon monoxide by way of carbon dioxide, set out as an equation with the intermediate shown
- the point of drawing it that way is that the enthalpy of the overall change does not depend on which route is taken to get there

Mean bond enthalpy

- stripping methane one hydrogen at a time takes 427, then 439, then 452, then 347 kJ mol-1, so the atomization of CH4 to C gas and 4H gas takes 1665 kJ mol-1
- in such cases we use the mean bond enthalpy of the C-H bond, which for methane is one quarter of 1665, that is 416 kJ mol-1
- mean C-H bond enthalpies differ slightly from compound to compound, as in CH3CH2Cl and CH3NO2, but not by a great deal, and using Hess's law bond enthalpies can be calculated

Reaction enthalpy from bond enthalpies

- reaction enthalpies are very important quantities, since they arise from the changes that accompany the breaking of old bonds and the formation of new ones, so we can predict the enthalpy of a gas phase reaction if we know the different bond enthalpies
- the standard enthalpy of reaction is the sum of the bond enthalpies of the reactants minus the sum of the bond enthalpies of the products, which is particularly useful when the required formation enthalpies are not available
- the net enthalpy change is the amount of energy required to break all the bonds in the reactant molecules minus the amount required to break all the bonds in the product molecules, and the relationship is approximate

Table 5.A Some Mean Single Bond Enthalpies in kJ mol–1 at 298 K

	H	C	N	O	F	Si	P	S	Cl	Br	I
H	435.8	414	389	464	569	293	318	339	431	368	297
C		347	293	351	439	289	264	259	330	276	238
N			159	201	272	368	209		201	243	201
O				138	184	540	351	327	205	197	213
F					155	176	490	226	255	289	213
Si							213	230	360	272	209
P							213	213	331	213	180
S									251	218	151
Cl									243	192	
Br											
I											

Table 5.B Some Mean Multiple Bond Enthalpies in kJ mol–1 at 298 K

Bond	kJ mol–1	Bond	kJ mol–1	Bond	kJ mol–1
N=N	418	C = C	611	O = O	498
N≡N	946	C ≡ C	837		
C=N	615	C = O	741		
C≡N	891	C ≡ O	1070		

Sodium metal and chlorine gas at the foot, sodium chloride at the top

Five steps between them, sublimation, ionization, dissociation, electron gain and lattice formation

The sum of the enthalpy changes round the closed cycle is zero, which is what makes the lattice enthalpy findable

Figure 5.M Enthalpy diagram for lattice enthalpy of NaCl

What lattice enthalpy is, and how it is found

- the lattice enthalpy of an ionic compound is the enthalpy change which occurs when one mole of it dissociates into its ions in the gaseous state, that of sodium chloride being plus 788 kJ mol-1
- since it is impossible to determine lattice enthalpies directly by experiment, we use an indirect method in which we construct an enthalpy diagram called a Born-Haber cycle
- the steps for sodium chloride are the sublimation of sodium metal at 108.4 kJ mol-1, the ionization of sodium atoms at 496 kJ mol-1, the dissociation of chlorine, whose reaction enthalpy is half the bond dissociation enthalpy at 121 kJ mol-1, the electron gained by chlorine atoms at minus 348.6 kJ mol-1, and finally the formation of the solid from the gaseous ions

Ionization energy against ionization enthalpy

- ionization energy and electron affinity are defined at absolute zero, and at any other temperature the heat capacities for the reactants and products have to be taken into account
- the value of Cp for each species is five halves R, so the change is plus five halves R for ionization and minus five halves R for electron gain
- therefore the ionization enthalpy is the ionization energy plus five halves RT, and the electron gain enthalpy is minus the electron affinity minus five halves RT

Closing the Born-Haber cycle

- the importance of the cycle is that the sum of the enthalpy changes round it is zero, so applying Hess's law the lattice enthalpy is 411.2 plus 108.4 plus 121 plus 496 minus 348.6, which gives plus 788 kJ for NaCl solid going to its gaseous ions
- the internal energy is smaller by 2RT, since Δng is 2, and equals plus 783 kJ mol-1
- using the lattice enthalpy of plus 788 kJ mol-1 and the hydration enthalpy of minus 784 kJ mol-1 from the literature, the enthalpy of solution comes to plus 4 kJ mol-1, so the dissolution of solid NaCl is accompanied by very little heat change

Enthalpy of solution

- the enthalpy of solution of a substance is the enthalpy change when one mole of it dissolves in a specified amount of solvent, and the enthalpy of solution at infinite dilution is the change observed on dissolving it in an infinite amount of solvent, where the interactions between the ions or solute molecules are negligible
- when an ionic compound dissolves the ions leave their ordered positions on the crystal lattice and become more free in solution, but solvation of those ions, hydration when the solvent is water, also occurs at the same time
- so the enthalpy of solution is determined by the lattice enthalpy and the enthalpy of hydration of the ions together, being the sum of the two

Why most salts dissolve better when warmed

- for most ionic compounds the enthalpy of solution is positive and the dissociation process is endothermic, so the solubility of most salts in water increases with a rise of temperature
- if the lattice enthalpy is very high the dissolution of the compound may not take place at all, which bears on why many fluorides tend to be less soluble than the corresponding chlorides
- estimates of the magnitudes of these enthalpy changes may be made by using tables of bond energies and lattice energies

Enthalpy of dilution

- the enthalpy of solution is the enthalpy change associated with adding a specified amount of solute to a specified amount of solvent at constant temperature and pressure
- dissolving one mole of gaseous hydrogen chloride in 10 mol of water has an enthalpy change of minus 69.01 kJ per mol, in 25 mol minus 72.03, in 40 mol minus 72.79, and in an infinite amount minus 74.85
- these values show the general dependence of the enthalpy of solution on the amount of solvent, so as more and more solvent is used the enthalpy of solution approaches a limiting value, the value in infinitely dilute solution

5.6 Spontaneity

What spontaneous means

- a gas does not spontaneously contract into one corner, and carbon dioxide will not form carbon and dioxygen spontaneously, so these and many other spontaneously occurring changes show a unidirectional change, which raises the question of what the driving force is and what determines the direction
- you may think a spontaneous reaction is one that occurs immediately when contact is made between the reactants, but hydrogen and oxygen may be mixed at room temperature and left for many years without any perceptible change, the reaction proceeding at an extremely slow rate and still being called spontaneous
- so spontaneity means having the potential to proceed without the assistance of an external agency, and it says nothing about the rate, while another aspect is that such changes cannot reverse their direction on their own, so a spontaneous process is an irreversible process and may only be reversed by some external agency

Is a decrease in enthalpy the criterion

- examining the flow of water downhill or the fall of a stone to the ground, there is a net decrease in potential energy in the direction of change, so by analogy we may be tempted to state that a chemical reaction is spontaneous in a given direction because a decrease in energy has taken place, as in exothermic reactions
- the formation of ammonia from its elements has a reaction enthalpy of minus 46.1 kJ mol-1, and the formation of hydrogen chloride is likewise exothermic

Reactants above products, the difference given out as heat

The change runs downhill in enthalpy

Figure 5.N Enthalpy diagram for exothermic reactions

Products above reactants, the difference taken in as heat

The change runs uphill in enthalpy and still happens on its own

Figure 5.O Enthalpy diagram for endothermic reactions

Two gases either side of a movable partition

The partition withdrawn, and the gases beginning to mix

After a time the mixing is complete, and it never unmixes

Figure 5.P Diffusion of two gases

Where the enthalpy criterion breaks down

- the postulate that the driving force for a chemical reaction may be a decrease in energy sounds reasonable on the evidence of exothermic reactions alone
- but some endothermic reactions are spontaneous as well, and the increase in enthalpy in those cases shows that a fall in enthalpy cannot be the whole criterion for spontaneity
- so we look instead at the diffusion of two gases separated by a movable partition, which once withdrawn lets them mix completely and never unmix, a change that happens on its own without any heat being given out

Entropy as disorder

- one can think of entropy as a measure of the degree of randomness or disorder in the system, so the greater the disorder in an isolated system the higher its entropy
- for a chemical reaction the entropy change can be attributed to the rearrangement of atoms or ions from one pattern in the reactants to another in the products, so if the structure of the products is much more disordered than that of the reactants there will be a resultant increase in entropy, and a decrease of regularity in structure means an increase in entropy
- for a given substance the crystalline solid state is the state of lowest entropy, that is the most ordered, and the gaseous state is the state of highest entropy

Entropy as a thermodynamic quantity

- entropy, like internal energy and enthalpy, is a state function, so ΔS is independent of path
- whenever heat is added to a system it increases molecular motions and so the randomness of the system, but the distribution of heat also depends on the temperature at which it is added, a system at higher temperature already having greater randomness, so temperature is the measure of the average chaotic motion of the particles
- heat added at lower temperature causes greater randomness than the same quantity added at higher temperature, which suggests that the entropy change is inversely proportional to temperature, so for a reversible reaction ΔS equals the reversible heat divided by T

Total entropy change, and what it tells apart

- the total entropy change for the system and surroundings of a spontaneous process is greater than zero, and when a system is in equilibrium the entropy is maximum and the change in entropy is zero
- so entropy for a spontaneous process increases till it reaches a maximum, and at equilibrium the change is zero
- for both reversible and irreversible expansion of an ideal gas under isothermal conditions ΔU is zero, but the total entropy change is not zero for the irreversible process, so ΔU does not discriminate between reversible and irreversible processes whereas ΔS does

Problem 5.10, does entropy rise or fall

- when a liquid crystallizes into a solid the molecules attain an ordered state, so entropy decreases
- at 0 K the constituent particles of a crystalline solid are static and entropy is minimum, so raising the temperature to 115 K sets them oscillating about their equilibrium positions in the lattice and the system becomes more disordered, so entropy increases
- in the decomposition of solid NaHCO3 the reactant is a solid with low entropy while the products are one solid and two gases, so the products represent a condition of higher entropy
- when one hydrogen molecule gives two atoms the number of particles increases, leading to a more disordered state, so two moles of H atoms have higher entropy than one mole of dihydrogen

Problem 5.11, iron rusting against its own entropy

- for the oxidation of iron to Fe2O3 the entropy change is minus 549.4 J K-1 mol-1 at 298 K, and the question asks why the reaction is spontaneous in spite of that, given a reaction enthalpy of minus 1648 × 103 J mol-1
- spontaneity is decided by the total entropy change, system plus surroundings, and the heat given out to the surroundings raises their entropy by more than the system's falls
- the total works out to 4980.6 J K-1 mol-1, which shows that the reaction is spontaneous

Gibbs energy and spontaneity

- for a system it is the total entropy change that decides the spontaneity of a process, but most chemical reactions fall into the category of either closed or open systems
- so for most chemical reactions there are changes in both enthalpy and entropy, and neither a decrease in enthalpy nor an increase in entropy alone can determine the direction of a spontaneous change

Deriving the criterion

- the total entropy change is that of the system plus that of the surroundings, and if the system is in thermal equilibrium with the surroundings then their temperatures are the same, while the increase in enthalpy of the surroundings equals the decrease in enthalpy of the system, so the entropy change of the surroundings is minus the system's enthalpy change over T
- rearranging gives T times the total entropy change equal to TΔSsys minus ΔHsys, and for a spontaneous process the total entropy change is positive, so ΔHsys minus TΔSsys is negative
- that quantity is ΔG, so ΔG = ΔH minus TΔS, and ΔHsys is the enthalpy change of the reaction while TΔSsys is the energy not available to do useful work, making ΔG the net energy available to do useful work and thus a measure of the free energy, which is why it is also known as the free energy of the reaction

The criterion, and the role of temperature

- ΔG gives a criterion for spontaneity at constant pressure and temperature, the process being spontaneous if ΔG is negative and non spontaneous if it is positive
- if a reaction has a positive enthalpy change and a positive entropy change it can still be spontaneous when TΔS is large enough to outweigh ΔH, which can happen in two ways, either the positive entropy change of the system is small and T must be large, or the entropy change is large and T may be small
- the former is one of the reasons why reactions are often carried out at high temperature

Entropy and the second law

- for an isolated system the change in energy remains constant, so an increase in entropy in such systems is the natural direction of a spontaneous change, and this is the second law of thermodynamics
- like the first law, the second can be stated in several ways, and it explains why spontaneous exothermic reactions are so common, since the heat released by the reaction increases the disorder of the surroundings and the overall entropy change is positive

Absolute entropy and the third law

- molecules of a substance may move in a straight line in any direction, they may spin like a top, and the bonds may stretch and compress, these being translational, rotational and vibrational motion, and when the temperature rises these motions become more vigorous and entropy increases, while when temperature is lowered entropy decreases
- the entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero, which is the third law of thermodynamics, and this is so because there is perfect order in a crystal at absolute zero
- the statement is confined to pure crystalline solids because theoretical arguments and practical evidence have shown that the entropy of solutions and super cooled liquids is not zero at 0 K, and the importance of the third law is that it permits the calculation of absolute values of entropy from thermal data alone, by summing the increments of reversible heat over temperature from 0 K to 298 K

5.7 Gibbs energy change and equilibrium

What free energy lets us predict

- a knowledge of the sign and magnitude of the free energy change of a chemical reaction allows prediction of the spontaneity of the reaction, and prediction of the useful work that could be extracted from it
- reversible in the strict thermodynamic sense is a special way of carrying out a process such that the system is at all times in perfect equilibrium with its surroundings
- applied to a chemical reaction, the term indicates that a given reaction can proceed in either direction simultaneously, so that a dynamic equilibrium is set up, which means the reactions in both directions should proceed with a decrease in free energy, which is only possible if at equilibrium the free energy of the system is at its lowest value

Table 5.C Effect of Temperature on Spontaneity of Reactions

$\Delta_r H$	$\Delta_r S$	$\Delta_r G$	Description*
–	+	–	Reaction spontaneous at all temperatures
–	–	– (at low T)	Reaction spontaneous at low temperature
–	–	+ (at high T)	Reaction nonspontaneous at high temperature
+	+	+ (at low T)	Reaction nonspontaneous at low temperature
+	+	– (at high T)	Reaction spontaneous at high temperature
+	–	+ (at all T)	Reaction nonspontaneous at all temperatures

What the relation is good for

- it is possible to obtain an estimate of the standard Gibbs energy change from measurements of ΔH and ΔS, and then calculate the equilibrium constant at any temperature for economic yields of the products
- and if the equilibrium constant is measured directly in the laboratory, the value of the standard Gibbs energy change at any other temperature can be calculated

Problems 5.12 and 5.13

- Problem 5.12 asks for the standard Gibbs energy change for the conversion of oxygen to ozone at 298 K, given an equilibrium constant of 2.47 × 10-29, and since it equals minus 2.303 RT log Kp with R at 8.314 J K-1 mol-1, the answer is 163000 J mol-1, that is 163 kJ mol-1
- Problem 5.13 asks for the equilibrium constant at 298 K when the standard Gibbs energy change is minus 13.6 kJ mol-1, and taking logs gives 2.38, so the equilibrium constant is the antilog of that, which is 2.4 × 102

Problem 5.14, dinitrogen tetroxide half dissociated

- at 60 degrees C dinitrogen tetroxide is 50 per cent dissociated, and the question asks for the standard free energy change at that temperature and at one atmosphere
- with N2O4 half dissociated the mole fractions follow from 1 minus 0.5 over 1 plus 0.5 for N2O4 and twice 0.5 over 1 plus 0.5 for NO2, so the partial pressures are 0.5 over 1.5 and 1 over 1.5 atm
- the equilibrium constant is the square of the NO2 pressure over the N2O4 pressure, which comes to 1.33 atm, and the standard free energy change follows from minus RT ln Kp

Summary

The first law, heat and work

- thermodynamics deals with energy changes in chemical or physical processes and enables us to study these changes quantitatively and to make useful predictions, and for these purposes we divide the universe into the system and the surroundings
- chemical or physical processes lead to evolution or absorption of heat, part of which may be converted into work, these quantities being related through the first law as $\Delta U = q + w$, where ΔU depends on initial and final states only and is a state function whereas q and w depend on the path and are not state functions
- we follow sign conventions by giving the positive sign to q and w when they are added to the system, and we can measure the transfer of heat from one system to another which causes a change in temperature, the magnitude of the rise depending on the heat capacity, so that the heat absorbed or evolved is $C\Delta T$
- work can be measured as minus the external pressure times the volume change in the case of expansion of gases, and under a reversible process we can put the external pressure equal to the internal one for infinitesimal changes, in which condition the gas equation applies

Enthalpy, and the enthalpy changes

- at constant volume w is zero and ΔU equals the heat transferred, but in the study of chemical reactions we usually have constant pressure, so we define another state function, enthalpy, whose change is $\Delta U + \Delta ng\ RT$ and can be found directly from the heat changes at constant pressure
- there are varieties of enthalpy change, and changes of phase such as melting, vaporization and sublimation usually occur at constant temperature and are characterized by enthalpy changes which are always positive
- enthalpy of formation, of combustion and other enthalpy changes can be calculated using Hess's law, and enthalpy change for chemical reactions can be determined from the enthalpies of formation of the substances involved or from the bond enthalpies

Exercises

Exercises 5.1 to 5.5

- 5.1 Choose the correct answer. A thermodynamic state function is a quantity (i) used to determine heat changes (ii) whose value is independent of path (iii) used to determine pressure volume work (iv) whose value depends on temperature only.
- 5.2 For the process to occur under adiabatic conditions, the correct condition is: (i) $\Delta T = 0$ (ii) $\Delta p = 0$ (iii) $q = 0$ (iv) $w = 0$
- 5.3 The enthalpies of all elements in their standard states are: (i) unity (ii) zero (iii) < 0 (iv) different for each element
- 5.4 ΔU of combustion of methane is – X kJ mol–1. The value of ΔH is (i) $= \Delta U$ (ii) $> \Delta U$ (iii) $< \Delta U$ (iv) $= 0$
- 5.5 The enthalpy of combustion of methane, graphite and dihydrogen at 298 K are, –890.3 kJ mol–1, –393.5 kJ mol–1, and –285.8 kJ mol–1 respectively. Enthalpy of formation of $CH_4(g)$ will be (i) –74.8 kJ mol–1 (ii) –52.27 kJ mol–1 (iii) +74.8 kJ mol–1 (iv) +52.26 kJ mol–1.

Exercises 5.6 to 5.8

- 5.6 A reaction, A + B → C + D + q is found to have a positive entropy change. The reaction will be (i) possible at high temperature (ii) possible only at low temperature (iii) not possible at any temperature (v) possible at any temperature
- 5.7 In a process, 701 J of heat is absorbed by a system and 394 J of work is done by the system. What is the change in internal energy for the process?
- 5.8 The reaction of cyanamide, NH_2CN (s), with dioxygen was carried out in a bomb calorimeter, and ΔU was found to be –742.7 kJ mol–1 at 298 K. Calculate enthalpy change for the reaction at 298 K. $NH_2CN(g) + 3/2\ O_2(g) \rightarrow N_2(g) + CO_2(g) + H_2O(l)$

Exercises, continued

Exercises 5.9 to 5.15

- 5.9 Calculate the number of kJ of heat necessary to raise the temperature of 60.0 g of aluminium from 35°C to 55°C. Molar heat capacity of Al is 24 J mol–1 K–1.
- 5.10 Calculate the enthalpy change on freezing of 1.0 mol of water at 10.0°C to ice at –10.0°C. ΔfusH = 6.03 kJ mol–1 at 0°C. Cp [H2O(l)] = 75.3 J mol–1 K–1, Cp [H2O(s)] = 36.8 J mol–1 K–1
- 5.11 Enthalpy of combustion of carbon to CO2 is –393.5 kJ mol–1. Calculate the heat released upon formation of 35.2 g of CO2 from carbon and dioxygen gas.
- 5.12 Enthalpies of formation of CO(g), CO2(g), N2O(g) and N2O4(g) are –110, –393, 81 and 9.7 kJ mol–1 respectively. Find the value of ΔrH for the reaction: N2O4(g) + 3CO(g) → N2O(g) + 3CO2(g)
- 5.13 Given N2(g) + 3H2(g) → 2NH3(g); ΔrH = –92.4 kJ mol–1. What is the standard enthalpy of formation of NH3 gas?
- 5.14 Calculate the standard enthalpy of formation of CH3OH(l) from the following data: CH3OH (l) + 3/2 O2(g) → CO2(g) + 2H2O(l) ; ΔrH = –726 kJ mol–1; C(graphite) + O2(g) → CO2(g) ; ΔcH = –393 kJ mol–1; H2(g) + 1/2 O2(g) → H2O(l); ΔfH = –286 kJ mol–1.
- 5.15 Calculate the enthalpy change for the process CCl4(g) → C(g) + 4 Cl(g) and calculate bond enthalpy of C – Cl in CCl4(g). ΔvapH(CCl4) = 30.5 kJ mol–1. ΔfH (CCl4) = –135.5 kJ mol–1. ΔaH (C) = 715.0 kJ mol–1, where ΔaH is enthalpy of atomisation. ΔaH (Cl2) = 242 kJ mol–1

Exercises 5.16 to 5.19

- 5.16 For an isolated system, ΔU = 0, what will be ΔS ?
- 5.17 For the reaction at 298 K, 2A + B → C, ΔH = 400 kJ mol–1 and ΔS = 0.2 kJ K–1 mol–1. At what temperature will the reaction become spontaneous considering ΔH and ΔS to be constant over the temperature range.
- 5.18 For the reaction, 2 Cl(g) → Cl2(g), what are the signs of ΔH and ΔS ?
- 5.19 For the reaction 2 A(g) + B(g) → 2D(g), ΔU = –10.5 kJ and ΔS = –44.1 JK–1. Calculate ΔG for the reaction, and predict whether the reaction may occur spontaneously.

Exercises, concluded

Exercises 5.20 to 5.22

- 5.20 The equilibrium constant for a reaction is 10. What will be the value of ΔG ? R = 8.314 JK–1 mol–1, T = 300 K.
- 5.21 Comment on the thermodynamic stability of NO(g), given 1/2 N2(g) + 1/2 O2(g) → NO(g); ΔrH = 90 kJ mol–1 and NO(g) + 1/2 O2(g) → NO2(g): ΔrH = –74 kJ mol–1
- 5.22 Calculate the entropy change in surroundings when 1.00 mol of H2O(l) is formed under standard conditions. ΔfH = –286 kJ mol–1.